1)

The line that goes through $(0,0,0)$ and is perpendicular to the plane is

$$\mathbf{r} = \begin{pmatrix}0\\0\\0\end{pmatrix} + \lambda\begin{pmatrix}1\\2\\9\end{pmatrix}$$

Substitute the line into the plane equation:

$(\lambda) + 2(2\lambda) + 9(9\lambda) = 344$

$\Rightarrow \lambda + 4\lambda + 81\lambda = 344$

$\Rightarrow 86\lambda - 344 = 0$

$\Rightarrow \lambda = 4$

So the line intersects the plane at the point $\big(4, 2(4), 9(4)\big) = (4, 8, 36)$

The distance between $(0,0,0)$ and $(4,8,36)$ is:

$\sqrt{(4-0)^2 + (8-0)^2 + (36-0)^2} = \sqrt{1376} = 4\sqrt{86}$ units

Alternative Method:

$$\mathbf{n} = \begin{pmatrix}n_1\\n_2\\n_3\end{pmatrix} = \begin{pmatrix}1\\2\\9\end{pmatrix}$$

$$\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

$d = -344$

$$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|1\times0 + 2\times0 + 9\times0 - 344|}{\sqrt{1^2 + 2^2 + 9^2}} = \frac{344}{\sqrt{86}} = 4\sqrt{86} \text{ units}$$

2)

The line that goes through $(2,1,2)$ and is perpendicular to the plane is

$$\mathbf{r} = \begin{pmatrix}2\\1\\2\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\1\end{pmatrix}$$

Substitute the line into the plane equation:

$(2+\lambda)+(1+\lambda)+(2+\lambda)=3$

$\Rightarrow 3\lambda+5=3$

$\Rightarrow 3\lambda+2=0$

$\Rightarrow \lambda=-\frac{2}{3}$

So the line intersects the plane at the point

$\left(2+\left(-\frac{2}{3}\right),1+\left(-\frac{2}{3}\right),2+\left(-\frac{2}{3}\right)\right)=\left(\frac{4}{3},\frac{1}{3},\frac{4}{3}\right)$

The distance between $(2,1,2)$ and $\left(\frac{4}{3},\frac{1}{3},\frac{4}{3}\right)$ is:

$\sqrt{\left(\frac{4}{3}-2\right)^2+\left(\frac{1}{3}-1\right)^2+\left(\frac{4}{3}-2\right)^2}=\sqrt{\frac{4}{3}}=\frac{2\sqrt{3}}{3}$ units

Alternative Method:

$$\mathbf{n}=\begin{pmatrix}n_1\\n_2\\n_3\end{pmatrix}=\begin{pmatrix}1\\1\\1\end{pmatrix}$$

$$\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}=\begin{pmatrix}2\\1\\2\end{pmatrix}$$

$d=-3$

$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}}=\frac{|1\times2+1\times1+1\times2-3|}{\sqrt{1^2+1^2+1^2}}=\frac{2}{\sqrt{3}}=\frac{2\sqrt{3}}{3}$ units

3) a)

The line that goes through $(10, 3, 1)$ and is perpendicular to the plane is

$$\mathbf{r} = \begin{pmatrix} 10 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

Substitute the line into the plane equation:

$(10 + \lambda) - (3 - \lambda) - (1 - \lambda) = 3$

$\Rightarrow 10 + \lambda - 3 + \lambda - 1 + \lambda = 3$

$\Rightarrow 3\lambda + 3 = 0$

$\Rightarrow \lambda = -1$

So the line intersects the plane at the point

$\big(10 + (-1), 3 - (-1), 1 - (-1)\big) = (9, 4, 2)$

The distance between $(10, 3, 1)$ and $(9, 4, 2)$ is:

$\sqrt{(9 - 10)^2 + (4 - 3)^2 + (2 - 1)^2} = \sqrt{3}$ units

Alternative Method:

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 10 \\ 3 \\ 1 \end{pmatrix}$$

$d = -3$

$$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|1 \times 10 + (-1) \times 3 + (-1) \times 1 - 3|}{\sqrt{1^2 + (-1)^2 + (-1)^2}} = \frac{3}{\sqrt{3}} = \sqrt{3} \text{ units}$$

3) b)

The line that goes through $(-3, 2, 8)$ and is perpendicular to the plane is

$$\mathbf{r} = \begin{pmatrix} -3 \\ 2 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 5 \\ -8 \end{pmatrix}$$

Substitute the line into the plane equation:

$2(-3 + 2\lambda) + 5(2 + 5\lambda) - 8(8 - 8\lambda) = 6$

$\Rightarrow -6 + 4\lambda + 10 + 25\lambda - 64 + 64\lambda = 6$

$\Rightarrow 93\lambda - 66 = 0$

$\Rightarrow \lambda = \frac{22}{31}$

So the line intersects the plane at the point

$$\left(-3 + 2\left(\frac{22}{31}\right), 2 + 5\left(\frac{22}{31}\right), 8 - 8\left(\frac{22}{31}\right)\right) = \left(-\frac{49}{31}, \frac{172}{31}, \frac{72}{31}\right)$$

The distance between $(-3, 2, 8)$ and $\left(-\frac{49}{31}, \frac{172}{31}, \frac{72}{31}\right)$ is:

$$\sqrt{\left(-\frac{49}{31} - (-3)\right)^2 + \left(\frac{172}{31} - 2\right)^2 + \left(\frac{72}{31} - 8\right)^2} = \sqrt{\frac{1452}{31}} = \frac{22\sqrt{93}}{31} \text{ units}$$

Alternative Method:

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ -8 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ 8 \end{pmatrix}$$

$d = -6$

$$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|2\times(-3) + 5\times2 + (-8)\times8 - 6|}{\sqrt{2^2 + 5^2 + (-8)^2}} = \frac{66}{\sqrt{93}} = \frac{22\sqrt{93}}{31} \text{ units}$$

3) c)

The line that goes through $(-9, 0, 60)$ and is perpendicular to the plane is

$$\mathbf{r} = \begin{pmatrix} -9 \\ 0 \\ 60 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ -3 \\ 9 \end{pmatrix}$$

Substitute the line into the plane equation:

$-5(-9 - 5\lambda) - 3(-3\lambda) + 9(60 + 9\lambda) = 10$

$\Rightarrow 45 + 25\lambda + 9\lambda + 540 + 81\lambda = 10$

$\Rightarrow 115\lambda + 575 = 0$

$\Rightarrow \lambda = -5$

So the line intersects the plane at the point

$\big(-9 - 5(-5), 0 - 3(-5), 60 + 9(-5)\big) = (16, 15, 15)$

The distance between $(-9, 0, 60)$ and $(16, 15, 15)$ is:

$\sqrt{\big(16 - (-9)\big)^2 + (15 - 0)^2 + (15 - 60)^2} = \sqrt{2875} = 5\sqrt{115}$ units

Alternative Method:

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} -5 \\ -3 \\ 9 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} -9 \\ 0 \\ 60 \end{pmatrix}$$

$d = -10$

$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|(-5)\times(-9) + (-3)\times 0 + 9\times 60 - 10|}{\sqrt{(-5)^2 + (-3)^2 + 9^2}} = \frac{575}{\sqrt{115}} = 5\sqrt{115}$ units

3) d)

The line that goes through $(6, 6, -1)$ and is perpendicular to the plane is

$$\mathbf{r} = \begin{pmatrix} 6 \\ 6 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -1 \\ -2 \end{pmatrix}$$

Substitute the line into the plane equation:

$-(6 - \lambda) - 2(-1 - 2\lambda) = 1$

$\Rightarrow -6 + \lambda + 2 + 4\lambda = 1$

$\Rightarrow 5\lambda - 5 = 0$

$\Rightarrow \lambda = 1$

So the line intersects the plane at the point

$\big(6, 6 - (1), -1 - 2(1)\big) = (6,5,-3)$

The distance between $(6, 6, -1)$ and $(6,5,-3)$ is:

$\sqrt{(6-6)^2 + (5-6)^2 + \big(-3-(-1)\big)^2} = \sqrt{5}$ units

Alternative Method:

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ -2 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \\ -1 \end{pmatrix}$$

$d = -1$

$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|0\times6 + (-1)\times6 + (-2)\times(-1) - 1|}{\sqrt{0^2 + (-1)^2 + (-2)^2}} = \frac{5}{\sqrt{5}} = \sqrt{5}$ units

4)

The line that goes through $(-10, 20, -50)$ and is perpendicular to the plane is

$$\mathbf{r} = \begin{pmatrix} -10 \\ 20 \\ -50 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 1 \\ -15 \end{pmatrix}$$

Substitute the line into the plane equation:

$4(-10 + 4\lambda) + (20 + \lambda) - 15(-50 - 15\lambda) = 1698$

$\Rightarrow -40 + 16\lambda + 20 + \lambda + 750 + 225\lambda = 1698$

$\Rightarrow 242\lambda - 968 = 0$

$\Rightarrow \lambda = 4$

So the line intersects the plane at the point

$\big(-10 + 4(4), 20 + (4), -50 - 15(4)\big) = (6, 24, -110)$

The distance between $(-10, 20, -50)$ and $(6, 24, -110)$ is:

$\sqrt{\big(6 - (-10)\big)^2 + (24 - 20)^2 + \big(-110 - (-50)\big)^2} = \sqrt{3872} = 44\sqrt{2}$ units

Alternative Method:

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ -15 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} -10 \\ 20 \\ -50 \end{pmatrix}$$

$d = -1698$

$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|4 \times (-10) + 1 \times 20 + (-15) \times (-50) - 1698|}{\sqrt{4^2 + 1^2 + (-15)^2}} = \frac{968}{\sqrt{242}} = 44\sqrt{2}$ units

5) a)

$10 + 30\lambda = 20 - 80\mu \Rightarrow 30\lambda + 80\mu = 10$ (1)

$5 + \lambda = 5 - 3\mu \Rightarrow \lambda + 3\mu = 0$ (2)

$-10 + 5\lambda = 10 + 5\mu \Rightarrow 5\lambda - 5\mu = 20$ (3)

Solve (1) and (2): $\lambda = 3$ and $\mu = -1$

Check in (3):

$5(3) - 5(-1) = 20$ as required

So the lines intersect.

Substitute $\lambda = 3$ into l_1:

The lines intersect at $P(100, 8, 5)$

5) b)

The line that goes through $(100, 8, 5)$ and is perpendicular to the plane is

$$\mathbf{r} = \begin{pmatrix} 100 \\ 8 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$$

Substitute the line into the plane equation:

$(100 + \lambda) - 2(8 - 2\lambda) + 3(5 + 3\lambda) - 218 = 0$

$\Rightarrow 100 + \lambda - 16 + 4\lambda + 15 + 9\lambda - 218 = 0$

$\Rightarrow 14\lambda - 119 = 0$

$\Rightarrow \lambda = \frac{17}{2}$

So the line intersects the plane at the point

$\left(100 + \left(\frac{17}{2}\right), 8 - 2\left(\frac{17}{2}\right), 5 + 3\left(\frac{17}{2}\right)\right) = \left(\frac{217}{2}, -9, \frac{61}{2}\right)$

The distance between $(100, 8, 5)$ and $\left(\frac{217}{2}, -9, \frac{61}{2}\right)$ is:

$\sqrt{\left(\frac{217}{2} - 100\right)^2 + (-9 - 8)^2 + \left(\frac{61}{2} - 5\right)^2} = \sqrt{\frac{2023}{2}} = \frac{17\sqrt{14}}{2}$ units

Alternative Method:

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 100 \\ 8 \\ 5 \end{pmatrix}$$

$d = -218$

$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|1 \times 100 + (-2) \times 8 + 3 \times 5 - 218|}{\sqrt{1^2 + (-2)^2 + 3^2}} = \frac{119}{\sqrt{14}} = \frac{17\sqrt{14}}{2}$ units

6)

$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ -14 \end{pmatrix}$

$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 5+3\lambda \\ 28+5\lambda \\ 18+\lambda \end{pmatrix}$

$d = 39$

$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}} = \sqrt{209}$

$\Rightarrow \frac{|3(5+3\lambda)-2(28+5\lambda)-14(18+\lambda)+39|}{\sqrt{3^2+(-2)^2+(-14)^2}} = \sqrt{209}$

$\Rightarrow \frac{|15+9\lambda-56-10\lambda-252-14\lambda+39|}{\sqrt{209}} = \sqrt{209}$

$\Rightarrow |-254-15\lambda| = 209$

$\Rightarrow -254-15\lambda = \pm 209 \Rightarrow \lambda = \frac{\pm 209-254}{15} = -3$ or $-\frac{463}{15}$

If $\lambda = -3$ then Q has coordinates $(5+3(-3), 28+5(-3), 18+(-3))$

$= (-4, 13, 15)$

If $\lambda = -\frac{463}{15}$ then Q has coordinates $\left(5+3\left(-\frac{463}{15}\right), 28+5\left(-\frac{463}{15}\right), 18+\left(-\frac{463}{15}\right)\right)$

$= \left(-\frac{438}{5}, -\frac{379}{3}-\frac{193}{15}\right)$

Alternative Method:

Q must have coordinates $(5+3\lambda, 28+5\lambda, 18+\lambda)$

The line that goes through Q and is perpendicular to the plane is

$\mathbf{r} = \begin{pmatrix} 5+3\lambda \\ 28+5\lambda \\ 18+\lambda \end{pmatrix} + \mu\begin{pmatrix} 3 \\ -2 \\ -14 \end{pmatrix} \Rightarrow \mathbf{r} = \begin{pmatrix} 5 \\ 28 \\ 18 \end{pmatrix} + \lambda\begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} + \mu\begin{pmatrix} 3 \\ -2 \\ -14 \end{pmatrix}$ (which is actually a plane).

Substitute the line into the plane equation:

$3(5+3\lambda+3\mu)-2(28+5\lambda-2\mu)-14(18+\lambda-14\mu)+39 = 0$

$\Rightarrow 15+9\lambda+9\mu-56-10\lambda+4\mu-252-14\lambda+196\mu+39 = 0$

$\Rightarrow -15\lambda+209\mu-254 = 0$

$\Rightarrow \mu = \frac{254+15\lambda}{209}$

So the line intersects the plane at the point

$\left(5+3\lambda+3\left(\frac{254+15\lambda}{209}\right), 28+5\lambda-2\left(\frac{254+15\lambda}{209}\right), 18+\lambda-14\left(\frac{254+15\lambda}{209}\right)\right)$

$= \left(\frac{672\lambda+1807}{209}, \frac{1015\lambda+5344}{209}, \frac{206-\lambda}{209}\right)$

The distance between $(5+3\lambda, 28+5\lambda, 18+\lambda)$ and $\left(\frac{672\lambda+1807}{209}, \frac{1015\lambda+5344}{209}, \frac{206-\lambda}{209}\right)$ is $\sqrt{209}$ units:

$$\sqrt{\left(\frac{672\lambda+1807}{209} - (5+3\lambda)\right)^2 + \left(\frac{1015\lambda+5344}{209} - (28+5\lambda)\right)^2 + \left(\frac{206-\lambda}{209} - (18+\lambda)\right)^2} = \sqrt{209}$$

$$= \sqrt{\left(\frac{762+45\lambda}{209}\right)^2 + \left(\frac{-508-30\lambda}{209}\right)^2 + \left(\frac{-3556-210\lambda}{209}\right)^2} = \sqrt{209}$$

$$= \left(\frac{762+45\lambda}{209}\right)^2 + \left(\frac{-508-30\lambda}{209}\right)^2 + \left(\frac{-3556-210\lambda}{209}\right)^2 = 209$$

$$= (762+45\lambda)^2 + (-508-30\lambda)^2 + (-3556-210\lambda)^2 = 209^3$$

$$\Rightarrow 2025\lambda^2 + 68580\lambda + 580644 + 900\lambda^2 + 30480\lambda + 258064 + 44100\lambda^2 + 1493520\lambda + 12645136 = 209^3$$

$$\Rightarrow 47025\lambda^2 + 1592580\lambda + 4354515 = 0$$

$$\Rightarrow 15\lambda^2 + 508\lambda + 1389 = 0$$

$$\Rightarrow (\lambda+3)(15\lambda+463) = 0$$

$$\Rightarrow \lambda = -3 \text{ or } \lambda = -\frac{463}{15}$$

If $\lambda = -3$ then Q has coordinates $\big(5+3(-3), 28+5(-3), 18+(-3)\big)$

$$= (-4, 13, 15)$$

If $\lambda = -\frac{463}{15}$ then Q has coordinates $\left(5+3\left(-\frac{463}{15}\right), 28+5\left(-\frac{463}{15}\right), 18+\left(-\frac{463}{15}\right)\right)$

$$= \left(-\frac{438}{5}, -\frac{379}{3} - \frac{193}{15}\right)$$

7) a)

$\mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ 50 \end{pmatrix} = -4935 \Rightarrow 2x + y + 50z = -4935$

The line that goes through $(25, 25, 0)$ and is perpendicular to the plane is

$\mathbf{r} = \begin{pmatrix} 25 \\ 25 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 50 \end{pmatrix}$

Substitute the line into the plane equation:

$2(25 + 2\lambda) + (25 + \lambda) + 50(50\lambda) = -4935$

$\Rightarrow 50 + 4\lambda + 25 + \lambda + 2500\lambda + 4935 = 0$

$\Rightarrow 2505\lambda + 5010 = 0$

$\Rightarrow \lambda = -2$

So the line intersects the plane at the point

$\big(25 + 2(-2), 25 + (-2), 50(-2)\big) = (21, 23, -100)$

The distance between $(25, 25, 0)$ and $(21, 23, -100)$ is:

$\sqrt{(21-25)^2 + (23-25)^2 + (-100-0)^2} = \sqrt{10020} = 2\sqrt{2505}$ m

Alternative Method:

$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 50 \end{pmatrix}$

$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 25 \\ 25 \\ 0 \end{pmatrix}$

$d = 4935$

$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|2\times25 + 1\times25 + 50\times0 + 4935|}{\sqrt{2^2 + 1^2 + 50^2}} = \frac{5010}{\sqrt{2505}} = 2\sqrt{2505}$ m

7) b)

Substitute the line $\mathbf{r} = \begin{pmatrix} 25 \\ 25 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ \frac{11}{7} \\ 1 \end{pmatrix}$ into the equation of the plane:

$2(25 + 10\lambda) + \left(25 + \frac{11}{7}\lambda\right) + 50(\lambda) = -4935$

$\Rightarrow 50 + 20\lambda + 25 + \frac{11}{7}\lambda + 50\lambda + 4935 = 0$

$\Rightarrow \frac{501}{7}\lambda + 5010 = 0$

$\Rightarrow \lambda = -70$

So the point where the tunnel intersects the coal bed is

$\left(25 + 10(-70), 25 + \frac{11}{7}(-70), -70\right) = (-675, -85, -70)$

Distance to the tunnel entrance:

$= \sqrt{(-675 - 25)^2 + (-85 - 25)^2 + (-70)^2} = \sqrt{507000} = 130\sqrt{30}$ m

8) a)

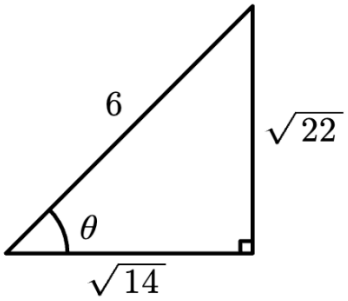
$\overrightarrow{OA} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \overrightarrow{OB} = \begin{pmatrix} 0 \\ -2 \\ -1 \end{pmatrix}, \overrightarrow{OC} = \begin{pmatrix} 2 \\ 2 \\ -2 \end{pmatrix}$

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$

$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{pmatrix} 2 \\ 2 \\ -2 \end{pmatrix} - \begin{pmatrix} 0 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix}$

$\cos\theta = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}||\overrightarrow{BC}|} = \frac{2\times2+1\times4+1\times(-1)}{\sqrt{2^2+1^2+1^2}\times\sqrt{2^2+4^2+(-1)^2}}$

$\Rightarrow \cos\theta = \frac{7}{\sqrt{6}\sqrt{21}} = \frac{\sqrt{14}}{6}$



$\therefore \sin\theta = \frac{\sqrt{22}}{6}$

8) b)

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ 2 & 4 & -1 \end{vmatrix} = (1 \times (-1) - 1 \times 4)\mathbf{i} - (2 \times (-1) - 1 \times 2)\mathbf{j} + (2 \times 4 - 1 \times 2)\mathbf{k}$$

$= -5\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$

$\therefore \Pi: -5x + 4y + 6z = d$

Substitute in $A(2, -1, 0)$:

$-5(2) + 4(-1) + 6(0) = d \Rightarrow d = -14$

$\therefore \Pi: -5x + 4y + 6z = -14$

$\Rightarrow -5x + 4y + 6z = -14$

8) c)

Substitute D into the plane:

$-5(3) + 4(1) + 6(5) = -15 + 4 + 30 = 19 \neq -14$

So D does not lie in the plane

8) d)

The line that goes through $(3,1,5)$ and is perpendicular to the plane is

$$\mathbf{r} = \begin{pmatrix}3\\1\\5\end{pmatrix} + \lambda\begin{pmatrix}-5\\4\\6\end{pmatrix}$$

Substitute the line into the plane equation:

$-5(3-5\lambda)+4(1+4\lambda)+6(5+6\lambda)=-14$

$\Rightarrow -15+25\lambda+4+16\lambda+30+36\lambda=-14$

$\Rightarrow 77\lambda+33=0$

$\Rightarrow \lambda=-\frac{3}{7}$

So the line intersects the plane at the point

$\left(3-5\left(-\frac{3}{7}\right),1+4\left(-\frac{3}{7}\right),5+6\left(-\frac{3}{7}\right)\right)=\left(\frac{36}{7},-\frac{5}{7},\frac{17}{7}\right)$

The distance between $(3,1,5)$ and $\left(\frac{36}{7},-\frac{5}{7},\frac{17}{7}\right)$ is:

$\sqrt{\left(\frac{36}{7}-3\right)^2+\left(-\frac{5}{7}-1\right)^2+\left(\frac{17}{7}-5\right)^2}=\sqrt{\frac{99}{7}}=\frac{3\sqrt{77}}{7}$ units

Alternative Method:

$$\mathbf{n}=\begin{pmatrix}n_1\\n_2\\n_3\end{pmatrix}=\begin{pmatrix}-5\\4\\6\end{pmatrix}$$

$$\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}=\begin{pmatrix}3\\1\\5\end{pmatrix}$$

$d=14$

$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}}=\frac{|(-5)\times3+4\times1+6\times5+14|}{\sqrt{(-5)^2+4^2+6^2}}=\frac{33}{\sqrt{77}}=\frac{3\sqrt{77}}{7}$ units

8) e)

Area of triangle $ABC=\frac{1}{2}\times\left|\overrightarrow{BA}\right|\left|\overrightarrow{BC}\right|\sin\theta$

$=\frac{1}{2}\times\sqrt{2^2+1^2+1^2}\times\sqrt{2^2+4^2+(-1)^2}\times\frac{\sqrt{22}}{6}$

$=\frac{1}{2}\times\sqrt{6}\times\sqrt{21}\times\frac{\sqrt{22}}{6}=\frac{\sqrt{77}}{2}$ units^2

Volume of tetrahedron $=\frac{1}{3}\times\frac{\sqrt{77}}{2}\times\frac{3\sqrt{77}}{7}=\frac{11}{2}$ units^3

9) a)

$\mathbf{r} \cdot \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = 5 \Rightarrow x + 3y + 2z = 5$

The line that goes through $(0, 0, 0)$ and is perpendicular to the plane is

$\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$

Substitute the line into the plane equation:

$(\lambda) + 3(3\lambda) + 2(2\lambda) = 5$

$\Rightarrow \lambda + 9\lambda + 4\lambda = 5$

$\Rightarrow 14\lambda - 5 = 0$

$\Rightarrow \lambda = \frac{5}{14}$

So the line intersects the plane at the point

$\left(\frac{5}{14}, 3\left(\frac{5}{14}\right), 2\left(\frac{5}{14}\right)\right) = \left(\frac{5}{14}, \frac{15}{14}, \frac{5}{7}\right)$

The distance between $(0, 0, 0)$ and $\left(\frac{5}{14}, \frac{15}{14}, \frac{5}{7}\right)$ is:

$\sqrt{\left(\frac{5}{14} - 0\right)^2 + \left(\frac{15}{14} - 0\right)^2 + \left(\frac{5}{7} - 0\right)^2} = \sqrt{\frac{25}{14}} = \frac{5\sqrt{14}}{14}$ units

Alternative Method:

$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$

$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$d = -5$

$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|1\times0 + 3\times0 + 2\times0 - 5|}{\sqrt{1^2 + 3^2 + 2^2}} = \frac{5}{\sqrt{14}} = \frac{5\sqrt{14}}{14}$ units

9) b)

Substitute the position vector $\begin{pmatrix} -13 \\ 4 \\ 3 \end{pmatrix}$ for $\mathbf{r}$:

$\begin{pmatrix} -13 \\ 4 \\ 3 \end{pmatrix}\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = (-13) \times 1 + 4 \times 3 + 3 \times 2 = -13 + 12 + 6 = 5$ as required

So $A(-13, 4, 3)$ lies on the plane Π

9) c)

$$\left|\begin{pmatrix}1\\3\\2\end{pmatrix}\right| = \sqrt{1^2+3^2+2^2} = \sqrt{14}$$

$$\hat{\mathbf{n}} = \frac{1}{\sqrt{14}}\begin{pmatrix}1\\3\\2\end{pmatrix} = \begin{pmatrix}\frac{1}{\sqrt{14}}\\ \frac{3}{\sqrt{14}}\\ \frac{2}{\sqrt{14}}\end{pmatrix}$$

So $\mathbf{r}\cdot\begin{pmatrix}\frac{1}{\sqrt{14}}\\ \frac{3}{\sqrt{14}}\\ \frac{2}{\sqrt{14}}\end{pmatrix} = d$

Find d by replacing $\mathbf{r}$ with the position vector of $A(-13,4,3)$:

$$\begin{pmatrix}-13\\4\\3\end{pmatrix}\cdot\begin{pmatrix}\frac{1}{\sqrt{14}}\\ \frac{3}{\sqrt{14}}\\ \frac{2}{\sqrt{14}}\end{pmatrix} = d$$

$$\Rightarrow d = (-13)\times\frac{1}{\sqrt{14}} + 4\times\frac{3}{\sqrt{14}} + 3\times\frac{2}{\sqrt{14}} = -\frac{13}{\sqrt{14}} + \frac{12}{\sqrt{14}} + \frac{6}{\sqrt{14}} = \frac{5}{\sqrt{14}} = \frac{5\sqrt{14}}{14}$$

So d is the distance from the origin to the plane

$$\therefore \mathbf{r}\cdot\begin{pmatrix}\frac{1}{\sqrt{14}}\\ \frac{3}{\sqrt{14}}\\ \frac{2}{\sqrt{14}}\end{pmatrix} = \frac{5\sqrt{14}}{14}$$

10) a)

$$\begin{pmatrix}2\\-3\\4\end{pmatrix}\cdot\begin{pmatrix}-2\\4\\5\end{pmatrix} = 2\times(-2) + (-3)\times 4 + 4\times 5 = 4$$

$$\therefore \Pi_2: \mathbf{r}\cdot\begin{pmatrix}-2\\4\\5\end{pmatrix} = 4$$

10) b)

$$\left|\begin{pmatrix}-2\\4\\5\end{pmatrix}\right| = \sqrt{(-2)^2+4^2+5^2} = \sqrt{45} = 3\sqrt{5}$$

$$\text{so } \hat{\mathbf{n}} = \frac{1}{3\sqrt{5}}\begin{pmatrix}-2\\4\\5\end{pmatrix} = \begin{pmatrix}-\frac{2}{3\sqrt{5}}\\\frac{4}{3\sqrt{5}}\\\frac{5}{3\sqrt{5}}\end{pmatrix} = \begin{pmatrix}-\frac{2\sqrt{5}}{15}\\\frac{4\sqrt{5}}{15}\\\frac{\sqrt{5}}{3}\end{pmatrix}$$

Find a single point that lies on Π_1:

$$\begin{pmatrix}k\\0\\0\end{pmatrix}\cdot\begin{pmatrix}-2\\4\\5\end{pmatrix} = 8 \Rightarrow k\times(-2)+0\times 4+0\times 5 = 8 \Rightarrow -2k = 8 \Rightarrow k = -4$$

So $(-4,0,0)$ lies on Π_1

$$\Pi_1: \mathbf{r}\cdot\begin{pmatrix}-\frac{2\sqrt{5}}{15}\\\frac{4\sqrt{5}}{15}\\\frac{\sqrt{5}}{3}\end{pmatrix} = d$$

Substitute the position vector of $(-4,0,0)$ for $\mathbf{r}$ to find d:

$$\begin{pmatrix}-4\\0\\0\end{pmatrix}\cdot\begin{pmatrix}-\frac{2\sqrt{5}}{15}\\\frac{4\sqrt{5}}{15}\\\frac{\sqrt{5}}{3}\end{pmatrix} = d \Rightarrow d = (-4)\times\left(-\frac{2\sqrt{5}}{15}\right)+0\times\frac{4\sqrt{5}}{15}+0\times\frac{\sqrt{5}}{3} = \frac{8\sqrt{5}}{15}$$

$$\therefore \Pi_1: \mathbf{r}\cdot\begin{pmatrix}-\frac{2\sqrt{5}}{15}\\\frac{4\sqrt{5}}{15}\\\frac{\sqrt{5}}{3}\end{pmatrix} = \frac{8\sqrt{5}}{15}$$

$$\Pi_2: \mathbf{r}\cdot\begin{pmatrix}-\frac{2\sqrt{5}}{15}\\\frac{4\sqrt{5}}{15}\\\frac{\sqrt{5}}{3}\end{pmatrix} = d$$

Substitute the position vector of $A(2,-3,4)$ for $\mathbf{r}$ to find d:

$$\begin{pmatrix}2\\-3\\4\end{pmatrix}\cdot\begin{pmatrix}-\frac{2\sqrt{5}}{15}\\\frac{4\sqrt{5}}{15}\\\frac{\sqrt{5}}{3}\end{pmatrix}=d\Rightarrow d=2\times\left(-\frac{2\sqrt{5}}{15}\right)+(-3)\times\frac{4\sqrt{5}}{15}+4\times\frac{\sqrt{5}}{3}$$

$$=-\frac{4\sqrt{5}}{15}-\frac{4\sqrt{5}}{5}+\frac{4\sqrt{5}}{3}=\frac{4\sqrt{5}}{15}$$

$$\therefore \Pi_2:\mathbf{r}\cdot\begin{pmatrix}-\frac{2\sqrt{5}}{15}\\\frac{4\sqrt{5}}{15}\\\frac{\sqrt{5}}{3}\end{pmatrix}=\frac{4\sqrt{5}}{15}$$

So $\Pi_1:\mathbf{r}\cdot\begin{pmatrix}-\frac{2\sqrt{5}}{15}\\\frac{4\sqrt{5}}{15}\\\frac{\sqrt{5}}{3}\end{pmatrix}=\frac{8\sqrt{5}}{15}$ and $\Pi_2:\mathbf{r}\cdot\begin{pmatrix}-\frac{2\sqrt{5}}{15}\\\frac{4\sqrt{5}}{15}\\\frac{\sqrt{5}}{3}\end{pmatrix}=\frac{4\sqrt{5}}{15}$

10) c)

$\frac{8\sqrt{5}}{15}$ is the distance from Π_1 to the origin, and $\frac{4\sqrt{5}}{15}$ is the distance from Π_2 to the origin. Both $\frac{8\sqrt{5}}{15}$ and $\frac{4\sqrt{5}}{15}$ are positive so both of the planes lie on the same side of the origin. So the distance from Π_1 to Π_2 (and hence from A to Π_1 as the question required), is $\frac{8\sqrt{5}}{15}-\frac{4\sqrt{5}}{15}=\frac{4\sqrt{5}}{15}$ units

11)

$\mathbf{r} \cdot \begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix} = 2 \Rightarrow 3x + 9y + z = 2$

The line that goes through $(2, -10, -642)$ and is perpendicular to the plane is

$\mathbf{r} = \begin{pmatrix} 2 \\ -10 \\ -642 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix}$

Substitute the line into the plane equation:

$3(2 + 3\lambda) + 9(-10 + 9\lambda) + (-642 + \lambda) = 2$

$\Rightarrow 6 + 9\lambda - 90 + 81\lambda - 642 + \lambda = 2$

$\Rightarrow 91\lambda - 728 = 0$

$\Rightarrow \lambda = 8$

So the line intersects the plane at the point

$\big(2 + 3(8), -10 + 9(8), -642 + (8)\big) = (26, 62, -634)$

The distance between $(2, -10, -642)$ and $(26, 62, -634)$ is:

$\sqrt{(26-2)^2 + \big(62 - (-10)\big)^2 + \big(-634 - (-642)\big)^2} = \sqrt{5824} = 8\sqrt{91}$ units

Alternative Method:

$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix}$

$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 2 \\ -10 \\ -642 \end{pmatrix}$

$d = -2$

$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|3\times2 + 9\times(-10) + 1\times(-642) - 2|}{\sqrt{3^2 + 9^2 + 1^2}} = \frac{728}{\sqrt{91}} = 8\sqrt{91}$ units

Alternative Method (using the method shown previously):

Let $\Pi_1 : \mathbf{r} \cdot \begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix} = 2$ and let Π_2 be the plane parallel to Π_1 that passes through A

$\begin{pmatrix} 0 \\ 0 \\ k \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix} = 2 \Rightarrow 0 \times 3 + 0 \times 9 + k \times 1 = 2 \Rightarrow k = 2$ so $(0, 0, 2)$ lies on Π_1

$\left| \begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix} \right| = \sqrt{3^2 + 9^2 + 1^2} = \sqrt{91}$

$$\hat{\mathbf{n}} = \frac{1}{\sqrt{91}}\begin{pmatrix}3\\9\\1\end{pmatrix} = \begin{pmatrix}\frac{3}{\sqrt{91}}\\ \frac{9}{\sqrt{91}}\\ \frac{1}{\sqrt{91}}\end{pmatrix}$$

$$\Pi_1: \mathbf{r} \cdot \begin{pmatrix}\frac{3}{\sqrt{91}}\\ \frac{9}{\sqrt{91}}\\ \frac{1}{\sqrt{91}}\end{pmatrix} = d$$

$$\begin{pmatrix}0\\0\\2\end{pmatrix} \cdot \begin{pmatrix}\frac{3}{\sqrt{91}}\\ \frac{9}{\sqrt{91}}\\ \frac{1}{\sqrt{91}}\end{pmatrix} = d \Rightarrow 0 \times \frac{3}{\sqrt{91}} + 0 \times \frac{9}{\sqrt{91}} + 2 \times \frac{1}{\sqrt{91}} = d \Rightarrow d = \frac{2}{\sqrt{91}}$$

$$\therefore \Pi_1: \mathbf{r} \cdot \begin{pmatrix}\frac{3}{\sqrt{91}}\\ \frac{9}{\sqrt{91}}\\ \frac{1}{\sqrt{91}}\end{pmatrix} = \frac{2}{\sqrt{91}}$$

$$\Pi_2: \mathbf{r} \cdot \begin{pmatrix}\frac{3}{\sqrt{91}}\\ \frac{9}{\sqrt{91}}\\ \frac{1}{\sqrt{91}}\end{pmatrix} = d$$

$$\begin{pmatrix}2\\-10\\-642\end{pmatrix} \cdot \begin{pmatrix}\frac{3}{\sqrt{91}}\\ \frac{9}{\sqrt{91}}\\ \frac{1}{\sqrt{91}}\end{pmatrix} = d \Rightarrow 2 \times \frac{3}{\sqrt{91}} + (-10) \times \frac{9}{\sqrt{91}} + (-642) \times \frac{1}{\sqrt{91}} = d \Rightarrow d = -\frac{726}{\sqrt{91}}$$

$$\therefore \Pi_2: \mathbf{r} \cdot \begin{pmatrix}\frac{3}{\sqrt{91}}\\ \frac{9}{\sqrt{91}}\\ \frac{1}{\sqrt{91}}\end{pmatrix} = -\frac{726}{\sqrt{91}}$$

$\frac{2}{\sqrt{91}}$ is the distance from Π_1 to the origin, and $\frac{726}{\sqrt{91}}$ is the distance from Π_2 to the origin. As $\frac{2}{\sqrt{91}}$ is positive and $-\frac{726}{\sqrt{91}}$ is negative, the planes lie on either side of the origin. So the distance from Π_1 to Π_2 (and hence from A to Π_1 as the question required), is $\frac{2}{\sqrt{91}} + \frac{726}{\sqrt{91}} = \frac{728}{\sqrt{91}} = 8\sqrt{91}$ units

12)

$$\begin{pmatrix}3\\5\\7\end{pmatrix}\times\begin{pmatrix}-2\\3\\9\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&5&7\\-2&3&9\end{vmatrix}$$

$$=(5\times9-7\times3)\mathbf{i}-\big(3\times9-7\times(-2)\big)\mathbf{j}+\big(3\times3-5\times(-2)\big)\mathbf{k}$$

$$=24\mathbf{i}-41\mathbf{j}+19\mathbf{k}$$

So the cartesian equation of the plane is $24x-41y+19z+d=0$

Substitute $(10,66,-8)$ into the plane to find d:

$24(10)-41(66)+19(-8)+d=0\Rightarrow d=2618$

$\therefore 24x-41y+19z+2618=0$

The line that goes through $(0,0,0)$ and is perpendicular to the plane is

$$\mathbf{r}=\begin{pmatrix}0\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}24\\-41\\19\end{pmatrix}$$

Substitute the line into the plane equation:

$24(24\lambda)-41(-41\lambda)+19(19\lambda)+2618=0$

$\Rightarrow 576\lambda+1681\lambda+361\lambda+2618=0$

$\Rightarrow 2618\lambda+2618=0$

$\Rightarrow \lambda=-1$

So the line intersects the plane at the point

$\big(24(-1),-41(-1),19(-1)\big)=(-24,41,-19)$

The distance between $(0,0,0)$ and $(-24,41,-19)$ is:

$\sqrt{(-24-0)^2+(41-0)^2+(-19-0)^2}=\sqrt{2618}$ units

Alternative Method:

$$\mathbf{n}=\begin{pmatrix}n_1\\n_2\\n_3\end{pmatrix}=\begin{pmatrix}24\\-41\\19\end{pmatrix}$$

$$\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

$d=2618$

$$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}}=\frac{|24\times0+(-41)\times0+19\times0+2618|}{\sqrt{24^2+(-41)^2+19^2}}=\frac{2618}{\sqrt{2618}}=\sqrt{2618}\text{ units}$$

13)

$$\begin{pmatrix}2\\-5\\1\end{pmatrix}\times\begin{pmatrix}-1\\1\\3\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&-5&1\\-1&1&3\end{vmatrix}$$

$$=\big((-5)\times 3-1\times 1\big)\mathbf{i}-\big(2\times 3-1\times(-1)\big)\mathbf{j}+\big(2\times 1-(-5)\times(-1)\big)\mathbf{k}$$

$$=-16\mathbf{i}-7\mathbf{j}-3\mathbf{k}$$

So the cartesian equation of the plane is $-16x-7y-3z+d=0$

Substitute $(120,-15,985)$ into the plane to find d:

$-16(120)-7(-15)-3(985)+d=0\Rightarrow d=4770$

$\therefore -16x-7y-3z+4770=0\Rightarrow 16x+7y+3z-4770=0$

The line that goes through $(2,1,7)$ and is perpendicular to the plane is

$$\mathbf{r}=\begin{pmatrix}2\\1\\7\end{pmatrix}+\lambda\begin{pmatrix}16\\7\\3\end{pmatrix}$$

Substitute the line into the plane equation:

$16(2+16\lambda)+7(1+7\lambda)+3(7+3\lambda)-4770=0$

$\Rightarrow 32+256\lambda+7+49\lambda+21+9\lambda-4770=0$

$\Rightarrow 314\lambda-4710=0$

$\Rightarrow \lambda=15$

So the line intersects the plane at the point

$\big(2+16(15),1+7(15),7+3(15)\big)=(242,106,52)$

The distance between $(2,1,7)$ and $(242,106,52)$ is:

$\sqrt{(242-2)^2+(106-1)^2+(52-7)^2}=\sqrt{70650}=15\sqrt{314}$ units

Alternative Method:

$$\mathbf{n}=\begin{pmatrix}n_1\\n_2\\n_3\end{pmatrix}=\begin{pmatrix}16\\7\\3\end{pmatrix}$$

$$\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}=\begin{pmatrix}2\\1\\7\end{pmatrix}$$

$d=-4770$

$$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}}=\frac{|16\times2+7\times1+3\times7-4770|}{\sqrt{16^2+7^2+3^2}}=\frac{4710}{\sqrt{314}}=15\sqrt{314}\text{ units}$$

14)

$\begin{pmatrix}2\\4\\1\end{pmatrix}-\begin{pmatrix}2\\3\\-5\end{pmatrix}=\begin{pmatrix}0\\1\\6\end{pmatrix}$

So the plane has equation $\mathbf{r}=\begin{pmatrix}2\\4\\1\end{pmatrix}+\lambda\begin{pmatrix}-1\\2\\4\end{pmatrix}+\mu\begin{pmatrix}0\\1\\6\end{pmatrix}$

$\begin{pmatrix}-1\\2\\4\end{pmatrix}\times\begin{pmatrix}0\\1\\6\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-1&2&4\\0&1&6\end{vmatrix}$

$=(2\times6-4\times1)\mathbf{i}-\big((-1)\times6-4\times0\big)\mathbf{j}+\big((-1)\times1-2\times0\big)\mathbf{k}$

$=8\mathbf{i}+6\mathbf{j}-\mathbf{k}$

So the cartesian equation of the plane is $8x+6y-z+d=0$

Substitute $(2,4,1)$ into the plane to find d:

$8(2)+6(4)-(1)+d=0\Rightarrow d=-39$

$\therefore 8x+6y-z-39=0$

The line that goes through $(0,0,0)$ and is perpendicular to the plane is

$\mathbf{r}=\begin{pmatrix}0\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}8\\6\\-1\end{pmatrix}$

Substitute the line into the plane equation:

$8(8\lambda)+6(6\lambda)-(-\lambda)-39=0$

$\Rightarrow 64\lambda+36\lambda+\lambda-39=0$

$\Rightarrow 101\lambda-39=0$

$\Rightarrow \lambda=\frac{39}{101}$

So the line intersects the plane at the point

$\left(8\left(\frac{39}{101}\right),6\left(\frac{39}{101}\right),-\left(\frac{39}{101}\right)\right)=\left(\frac{312}{101},\frac{234}{101},-\frac{39}{101}\right)$

The distance between $(0,0,0)$ and $\left(\frac{312}{101},\frac{234}{101},-\frac{39}{101}\right)$ is:

$\sqrt{\left(\frac{312}{101}-0\right)^2+\left(\frac{234}{101}-0\right)^2+\left(-\frac{39}{101}-0\right)^2}=\sqrt{\frac{1521}{101}}=\frac{39\sqrt{101}}{101}$ units

Alternative Method:

$\mathbf{n}=\begin{pmatrix}n_1\\n_2\\n_3\end{pmatrix}=\begin{pmatrix}8\\6\\-1\end{pmatrix}$

$\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$

$d = -39$

$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}} = \frac{|8\times0+6\times0+(-1)\times0-39|}{\sqrt{8^2+6^2+(-1)^2}} = \frac{39}{\sqrt{101}} = \frac{39\sqrt{101}}{101}$ units

15)

Let Π_1 be $x + y + z = 40$ and let Π_2 be $x + y + z = 60$

Choose a point on the plane Π_1: $(10, 10, 20)$

The line that goes through $(10, 10, 20)$ and is perpendicular to the plane Π_2 is

$$\mathbf{r} = \begin{pmatrix}10\\10\\20\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\1\end{pmatrix}$$

Substitute the line into the plane equation Π_2:

$(10 + \lambda) + (10 + \lambda) + (20 + \lambda) = 60$

$\Rightarrow 3\lambda - 20 = 0$

$\Rightarrow \lambda = \frac{20}{3}$

So the line intersects the plane Π_2 at the point

$\left(10 + \left(\frac{20}{3}\right), 10 + \left(\frac{20}{3}\right), 20 + \left(\frac{20}{3}\right)\right) = \left(\frac{50}{3}, \frac{50}{3}, \frac{80}{3}\right)$

The distance between $(10, 10, 20)$ and $\left(\frac{50}{3}, \frac{50}{3}, \frac{80}{3}\right)$ is:

$\sqrt{\left(\frac{50}{3} - 10\right)^2 + \left(\frac{50}{3} - 10\right)^2 + \left(\frac{80}{3} - 20\right)^2} = \sqrt{\frac{400}{3}} = \frac{20\sqrt{3}}{3}$ units

Alternative Method:

$$\mathbf{n} = \begin{pmatrix}n_1\\n_2\\n_3\end{pmatrix} = \begin{pmatrix}1\\1\\1\end{pmatrix}$$

$$\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix} = \begin{pmatrix}10\\10\\20\end{pmatrix}$$

$d = -60$

$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}} = \frac{|1\times10+1\times10+1\times20-60|}{\sqrt{1^2+1^2+1^2}} = \frac{20}{\sqrt{3}} = \frac{20\sqrt{3}}{3}$ units

16) a)

Substitute $(100, -150, -4)$ into Π_1:

$p(100) + 2(-150) + q(-4) = 300 \Rightarrow 100p - 4q = 600$ (1)

Substitute $(150, -100, 10)$ into Π_1:

$p(150) + 2(-100) + q(10) = 300 \Rightarrow 150p + 10q = 500$ (2)

Solve (1) and (2): $p = 5$ and $q = -25$

16) b)

The line that goes through $(100, -150, -4)$ and is perpendicular to the plane Π_2 is

$$\mathbf{r} = \begin{pmatrix} 100 \\ -150 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 2 \\ -25 \end{pmatrix}$$

Substitute the line into the plane equation Π_2:

$5(100 + 5\lambda) + 2(-150 + 2\lambda) - 25(-4 - 25\lambda) = 627$

$\Rightarrow 500 + 25\lambda - 300 + 4\lambda + 100 + 625\lambda = 627$

$\Rightarrow 654\lambda - 327 = 0$

$\Rightarrow \lambda = \frac{1}{2}$

So the line intersects the plane Π_2 at the point

$\left(100 + 5\left(\frac{1}{2}\right), -150 + 2\left(\frac{1}{2}\right), -4 - 25\left(\frac{1}{2}\right)\right) = \left(\frac{205}{2}, -149, -\frac{33}{2}\right)$

The distance between $(100, -150, -4)$ and $\left(\frac{205}{2}, -149, -\frac{33}{2}\right)$ is:

$$\sqrt{\left(\frac{205}{2} - 100\right)^2 + \left(-149 - (-150)\right)^2 + \left(-\frac{33}{2} - (-4)\right)^2} = \sqrt{\frac{327}{2}} = \frac{\sqrt{654}}{2} \text{ units}$$

Alternative Method:

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ -25 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 100 \\ -150 \\ -4 \end{pmatrix}$$

$d = -627$

$$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|5 \times 100 + 2 \times (-150) + (-25) \times (-4) - 627|}{\sqrt{5^2 + 2^2 + (-25)^2}} = \frac{327}{\sqrt{654}} = \frac{\sqrt{654}}{2} \text{ units}$$

17)

The normal vector of the plane Π_1 is $\begin{pmatrix}1\\2\\2\end{pmatrix}$

A vector parallel to $\begin{pmatrix}1\\2\\2\end{pmatrix}$ of length 5 units is $\frac{5}{\sqrt{1^2+2^2+2^2}}\begin{pmatrix}1\\2\\2\end{pmatrix}=\frac{5}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}=\begin{pmatrix}\frac{5}{3}\\\frac{10}{3}\\\frac{10}{3}\end{pmatrix}$

Choose a point on the plane Π_1: $(0,0,5)$

So the position vector of a point on the plane Π_2 is $\begin{pmatrix}0\\0\\5\end{pmatrix}\pm\begin{pmatrix}\frac{5}{3}\\\frac{10}{3}\\\frac{10}{3}\end{pmatrix}$

First consider $\begin{pmatrix}0\\0\\5\end{pmatrix}+\begin{pmatrix}\frac{5}{3}\\\frac{10}{3}\\\frac{10}{3}\end{pmatrix}=\begin{pmatrix}\frac{5}{3}\\\frac{10}{3}\\\frac{25}{3}\end{pmatrix}$

Substitute $\left(\frac{5}{3},\frac{10}{3},\frac{25}{3}\right)$ into Π_2:

$\left(\frac{5}{3}\right)+2\left(\frac{10}{3}\right)+2\left(\frac{25}{3}\right)+d_1=0\Rightarrow d_1=-25$

$\therefore \Pi_2: x+2y+2z-25=0$

Now consider $\begin{pmatrix}0\\0\\5\end{pmatrix}-\begin{pmatrix}\frac{5}{3}\\\frac{10}{3}\\\frac{10}{3}\end{pmatrix}=\begin{pmatrix}-\frac{5}{3}\\-\frac{10}{3}\\\frac{5}{3}\end{pmatrix}$

Substitute $\left(-\frac{5}{3},-\frac{10}{3},\frac{5}{3}\right)$ into Π_2:

$\left(-\frac{5}{3}\right)+2\left(-\frac{10}{3}\right)+2\left(\frac{5}{3}\right)+d_2=0\Rightarrow d_2=5$

$\therefore \Pi_2: x+2y+2z+5=0$

Alternative Method:

$\mathbf{n}=\begin{pmatrix}n_1\\n_2\\n_3\end{pmatrix}=\begin{pmatrix}1\\2\\2\end{pmatrix}$

$\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}=\begin{pmatrix}0\\0\\5\end{pmatrix}$

$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}} = 5$

$\Rightarrow \frac{|1\times0+2\times0+2\times5+d|}{\sqrt{1^2+2^2+2^2}} = 5$

$\Rightarrow |10+d| = 15$

$\Rightarrow 10+d = \pm 15$

$\Rightarrow d = -10 \pm 15 = 5 \text{ or } -25$

$\therefore \Pi_2: x+2y+2z+5=0 \text{ or } \Pi_2: x+2y+2z-25=0$

18) a)

Find the line passing through $(4,2,0)$ and $(4,10,8)$

$\begin{pmatrix}4\\10\\8\end{pmatrix} - \begin{pmatrix}4\\2\\0\end{pmatrix} = \begin{pmatrix}0\\8\\8\end{pmatrix}$

$\therefore \mathbf{r} = \begin{pmatrix}4\\2\\0\end{pmatrix} + \lambda\begin{pmatrix}0\\8\\8\end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix}4\\2\\0\end{pmatrix} + \lambda\begin{pmatrix}0\\1\\1\end{pmatrix}$

$\sin\theta = \frac{\sqrt{6}}{3} \Rightarrow \cos\theta = \frac{\sqrt{3}}{3}$

$\frac{\sqrt{3}}{3} = \frac{\begin{pmatrix}p\\9\\3\end{pmatrix}\cdot\begin{pmatrix}0\\1\\1\end{pmatrix}}{\sqrt{p^2+9^2+3^2}\times\sqrt{0^2+1^2+1^2}}$

$\Rightarrow \frac{\sqrt{3}}{3} = \frac{p\times0+9\times1+3\times1}{\sqrt{p^2+90}\times\sqrt{2}}$

$\Rightarrow \frac{\sqrt{6}}{3}\times\sqrt{p^2+90} = 12$

$\Rightarrow \sqrt{p^2+90} = 6\sqrt{6}$

$\Rightarrow p^2+90 = 216$

$\Rightarrow p^2 = 126$

$\Rightarrow p = \pm 3\sqrt{14}$

As $p > 0$, $p = 3\sqrt{14}$

18) b)

Substitute $\mathbf{r} = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ into the plane $3\sqrt{14}x + 9y + 3z = 6$ or $\sqrt{14}x + 3y + z = 2$:

$\sqrt{14}(4) + 3(2 + \lambda) + (\lambda) = 2$

$\Rightarrow 4\sqrt{14} + 6 + 3\lambda + \lambda = 2$

$\Rightarrow 4\lambda + 4\sqrt{14} + 4 = 0$

$\Rightarrow \lambda = -\sqrt{14} - 1$

So the line intersects the plane at $\left(4, 2 + \left(-\sqrt{14} - 1\right), -\sqrt{14} - 1\right)$

$= \left(4, 1 - \sqrt{14}, -1 - \sqrt{14}\right)$

18) c)

The line that goes through $\left(4, 1 - \sqrt{14}, -1 - \sqrt{14}\right)$ and is perpendicular to the plane Π_2 is

$$\mathbf{r} = \begin{pmatrix} 4 \\ 1 - \sqrt{14} \\ -1 - \sqrt{14} \end{pmatrix} + \lambda \begin{pmatrix} \sqrt{14} \\ 3 \\ 1 \end{pmatrix}$$

Substitute the line into the plane equation Π_2:

$\sqrt{14}\left(4 + \sqrt{14}\lambda\right) + 3\left(1 - \sqrt{14} + 3\lambda\right) + \left(-1 - \sqrt{14} + \lambda\right) = 26$

$\Rightarrow 4\sqrt{14} + 14\lambda + 3 - 3\sqrt{14} + 9\lambda - 1 - \sqrt{14} + \lambda = 26$

$\Rightarrow 24\lambda - 24 = 0$

$\Rightarrow \lambda = 1$

So the line intersects the plane Π_2 at the point

$\left(4 + \sqrt{14}(1), 1 - \sqrt{14} + 3(1), -1 - \sqrt{14} + 1(1)\right) = \left(4 + \sqrt{14}, 4 - \sqrt{14}, -\sqrt{14}\right)$

The distance between $\left(4, 1 - \sqrt{14}, -1 - \sqrt{14}\right)$ and $\left(4 + \sqrt{14}, 4 - \sqrt{14}, -\sqrt{14}\right)$ is:

$$\sqrt{\left(4 + \sqrt{14} - 4\right)^2 + \left(4 - \sqrt{14} - \left(1 - \sqrt{14}\right)\right)^2 + \left(-\sqrt{14} - \left(-1 - \sqrt{14}\right)\right)^2}$$

$\sqrt{14 + 9 + 1} = \sqrt{24} = 2\sqrt{6}$ m

Alternative Method:

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} \sqrt{14} \\ 3 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1-\sqrt{14} \\ -1-\sqrt{14} \end{pmatrix}$$

$d = -26$

$$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}} = \frac{|\sqrt{14}\times 4+3\times(1-\sqrt{14})+1\times(-1-\sqrt{14})-2|}{\sqrt{(\sqrt{14})^2+3^2+1^2}} = \frac{24}{\sqrt{24}} = 2\sqrt{6}\text{ m}$$

19)

$\begin{pmatrix}3\\2\\5\end{pmatrix}\cdot\begin{pmatrix}-31\\14\\13\end{pmatrix}=3\times(-31)+2\times14+5\times13=0$

So the line is either parallel to the plane, or it lies in the plane.

Check whether $(2,8,-1)$ lies in the plane:

$-31(2)+14(8)+13(-1)+2615=2652\neq0$

So the point does not lie on the line.

The line that goes through $(2,8,-1)$ and is perpendicular to the plane is

$\mathbf{r}=\begin{pmatrix}2\\8\\-1\end{pmatrix}+\lambda\begin{pmatrix}-31\\14\\13\end{pmatrix}$

Substitute the line into the plane equation:

$-31(2-31\lambda)+14(8+14\lambda)+13(-1+13\lambda)+2615=0$

$\Rightarrow-62+961\lambda+112+196\lambda-13+169\lambda+2615=0$

$\Rightarrow1326\lambda+2652=0$

$\Rightarrow\lambda=-2$

So the line intersects the plane at the point

$\big(2-31(-2),8+14(-2),-1+13(-2)\big)=(64,-20,-27)$

The distance between $(2,8,-1)$ and $(64,-20,-27)$ is:

$\sqrt{(64-2)^2+(-20-8)^2+\big(-27-(-1)\big)^2}=\sqrt{5304}=2\sqrt{1326}$ units

Alternative Method:

$\mathbf{n}=\begin{pmatrix}n_1\\n_2\\n_3\end{pmatrix}=\begin{pmatrix}-31\\14\\13\end{pmatrix}$

$\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}=\begin{pmatrix}2\\8\\-1\end{pmatrix}$

$d=2615$

$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}}=\frac{|-31\times2+14\times8+13\times(-1)+2615|}{\sqrt{(-31)^2+14^2+13^2}}=\frac{2652}{\sqrt{1326}}=2\sqrt{1326}$ units

20)

$$2x = \frac{y-3}{2} = \frac{60-6z}{13} \Rightarrow \frac{x}{\frac{1}{2}} = \frac{y-3}{2} = \frac{z-10}{-\frac{13}{6}} \Rightarrow \mathbf{r} = \begin{pmatrix} 0 \\ 3 \\ 10 \end{pmatrix} + \lambda \begin{pmatrix} \frac{1}{2} \\ 2 \\ -\frac{13}{6} \end{pmatrix}$$

$$\mathbf{r} \cdot \begin{pmatrix} 3 \\ -4 \\ -3 \end{pmatrix} = 10 \Rightarrow 3x - 4y - 3z = 10$$

$$\begin{pmatrix} \frac{1}{2} \\ 2 \\ -\frac{13}{6} \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \\ -3 \end{pmatrix} = \frac{1}{2} \times 3 + 2 \times (-4) + \left(-\frac{13}{6}\right) \times (-3) = 0$$

So the line is either parallel to the plane, or it lies in the plane.

Check whether $(0, 3, 10)$ lies in the plane:

$3(0) - 4(3) - 3(10) = -42 \neq 10$

So the point does not lie on the line.

The line that goes through $(0, 3, 10)$ and is perpendicular to the plane is

$$\mathbf{r} = \begin{pmatrix} 0 \\ 3 \\ 10 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -4 \\ -3 \end{pmatrix}$$

Substitute the line into the plane equation:

$3(3\lambda) - 4(3 - 4\lambda) - 3(10 - 3\lambda) = 60$

$\Rightarrow 9\lambda - 12 + 16\lambda - 30 + 9\lambda = 60$

$\Rightarrow 34\lambda - 102 = 0$

$\Rightarrow \lambda = 3$

So the line intersects the plane at the point

$\big(0 + 3(3), 3 - 4(3), 10 - 3(3)\big) = (9, -9, 1)$

The distance between $(0, 3, 10)$ and $(9, -9, 1)$ is:

$\sqrt{(9-0)^2 + (-9-3)^2 + (1-10)^2} = \sqrt{306} = 3\sqrt{34}$ units

Alternative Method:

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 10 \end{pmatrix}$$

$d = -60$

$$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}} = \frac{|3\times0+(-4)\times3+(-3)\times10-60|}{\sqrt{3^2+(-4)^2+(-3)^2}} = \frac{102}{\sqrt{34}} = 3\sqrt{34} \text{ units}$$

21)

First, find the shortest distance from the point to the plane.

$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 8 \end{pmatrix}$

$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$

$d = 65$

$\frac{|n_1x_1+n_2y_1+n_3z_1+d|}{\sqrt{n_1^2+n_2^2+n_3^2}} = \frac{|4\times2+1\times1+8\times2+65|}{\sqrt{4^2+1^2+8^2}} = \frac{90}{\sqrt{81}} = 10$

Now we need to set the shortest from the point to the line to be 10 to find k:

Let $\mathbf{p} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ -13 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 4 \\ 3 \\ k \end{pmatrix}$

Then $10 = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 4 \\ 3 \\ k \end{pmatrix}\times\left(\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}-\begin{pmatrix} 0 \\ -13 \\ 0 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 4 \\ 3 \\ k \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 4 \\ 3 \\ k \end{pmatrix}\times\begin{pmatrix} 2 \\ 14 \\ 2 \end{pmatrix}\right|}{\left|\begin{pmatrix} 4 \\ 3 \\ k \end{pmatrix}\right|}$

$\begin{pmatrix} 4 \\ 3 \\ k \end{pmatrix}\times\begin{pmatrix} 2 \\ 14 \\ 2 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 3 & k \\ 2 & 14 & 2 \end{vmatrix}$

$= (3\times2-k\times14)\mathbf{i}-(4\times2-k\times2)\mathbf{j}+(4\times14-3\times2)\mathbf{k}$

$= (6-14k)\mathbf{i}+(2k-8)\mathbf{j}+50\mathbf{k}$

$\therefore 10 = \frac{\left|\begin{pmatrix} 6-14k \\ 2k-8 \\ 50 \end{pmatrix}\right|}{\left|\begin{pmatrix} 4 \\ 3 \\ k \end{pmatrix}\right|} = \frac{\sqrt{(6-14k)^2+(2k-8)^2+50^2}}{\sqrt{4^2+3^2+k^2}}$

$\Rightarrow 10 = \frac{\sqrt{196k^2-168k+36+4k^2-32k+64+2500}}{\sqrt{25+k^2}}$

$\Rightarrow 100 = \frac{200k^2-200k+2600}{25+k^2}$

$\Rightarrow 1 = \frac{2k^2-2k+26}{25+k^2}$

$\Rightarrow 25+k^2 = 2k^2-2k+26$

$\Rightarrow 0 = k^2-2k+1$

$\Rightarrow 0 = (k-1)^2 \Rightarrow k = 1$